

# Workflow Overview

The recommendation system follows a **multi-stage evidence-fusion pipeline**:

## 1. Free-text Research Query

Researchers provide a natural language clinical request (for example: *Identify relevant codes for Type 2 Diabetes*).

## 2. Keyword Extraction

The query is converted into structured search terms for downstream retrieval.

## 3. Codelist Discovery

Candidate codelists are retrieved using multiple similarity methods:

- TF-IDF similarity
- SapBERT embeddings
- MiniLM embeddings

This prioritises clinically meaningful grouped concepts.

## 4. Semantic Retrieval

SapBERT embeddings with FAISS vector search retrieve semantically related clinical concepts.

## 5. Hierarchical Expansion

SNOMED CT hierarchy traversal using Lowest Common Ancestor (LCA) identifies structurally related concepts.

## 6. Validation Layer

Results are validated against:

- NHS reference sets
- QOF clinical rules

Confidence is upgraded/downgraded based on evidence agreement.

## 7. LLM Justification Layer

Google Gemini 2.5 Flash-Lite provides inclusion reasoning, confidence explanation, and contextual justification.

## 8. Ranked Output Generation

The final output is a structured Excel recommendation pack with:

- Full recommendation list
- High confidence list
- Medium confidence list
- Low confidence list
- Evidence trace columns
- Justification / reasoning column